

Avian Functional Trait Database Explorer

Objective

Design and implement a reproducible data ingestion, analysis, and visualisation pipeline to explore large avian trait databases, with the goal of understanding ecomorphological relationships, data quality issues, and variability across avian taxa.

Core Task

- Identify and ingest **public bird trait datasets** (e.g., body mass, wing length, tail length, hand-wing index, or beak properties) from open databases.
- Design a structured data model to store taxonomic identifiers, morphological descriptors, ecological properties, units, and source metadata.
- Clean and harmonise trait data, addressing inconsistent units, missing values, duplicated entries, and taxonomic synonyms across sources.
- Compute summary statistics and exploratory relationships between functional traits and ecological roles.
- Build an interactive dashboard that enables filtering, comparison, and visual exploration of trait distributions and phylogenetic correlations.

Intensification Requirements

- Compare **at least two strategies** for handling missing or conflicting property measurements and analyse their impact on inferred trends.
- Analyse how data coverage and quality vary across taxonomic orders or biogeographical realms.
- Explicitly discuss **measurement uncertainty and source reliability**, including differences between observed and estimated values.
- Include at least one **iterative refinement** where data cleaning rules, descriptor selection, or visual encodings are revised based on exploratory findings.
- Assess scalability and performance as the number of species or traits increases.

Expected Outcome

A well-documented avian traits exploration system that demonstrates principled data curation, exploratory analysis, and thoughtful interpretation of form–function relationships in real-world biological datasets.